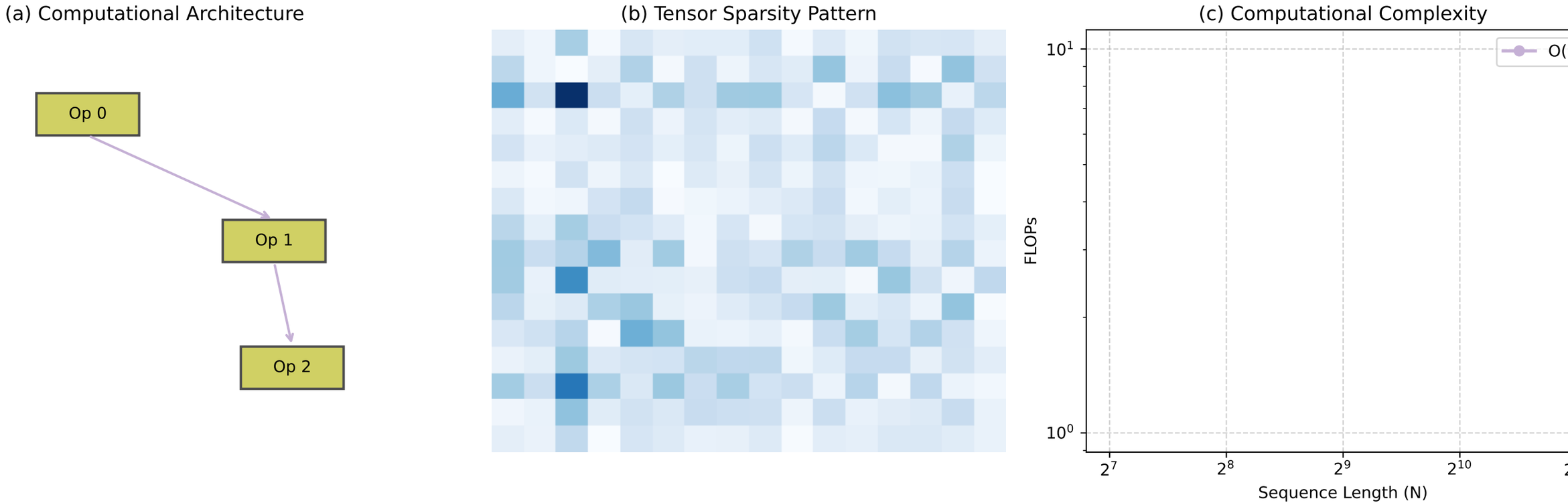Module Architecture: checkpoint